

MATB34 - Written Assignment

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1 Laplace transforms

(1)

$$\mathcal{L}\{\cos^2(\omega t)\} = \mathcal{L}\left\{\frac{1}{2}(1 + \cos(2\omega t))\right\} = \frac{1}{2}(\mathcal{L}\{1\} + \mathcal{L}\{\cos(2\omega t)\}) = \frac{1}{2}\left(\frac{1}{s} + \frac{s}{s^2 + 4\omega^2}\right).$$

(2)

$$\mathcal{L}\{t^2 - 1\} = \mathcal{L}\{t^2\} - \mathcal{L}\{1\} = \frac{2}{s^3} - \frac{1}{s}.$$

(3)

$$\mathcal{L}\{e^{-t} \cos^2 t\} = \mathcal{L}\left\{e^{-t} \cdot \frac{1}{2}(1 + \cos(2t))\right\} = \frac{1}{2}\mathcal{L}\{e^{-t}\} + \frac{1}{2}\mathcal{L}\{e^{-t} \cos(2t)\} = \frac{1}{2(s+1)} + \frac{s+1}{2((s+1)^2 + 4)}.$$

2 Inverse Laplace transforms

(1)

$$\begin{aligned}\mathcal{L}^{-1}\left\{\frac{1}{(s-a)(s-b)}\right\} &= \mathcal{L}^{-1}\left\{\frac{1}{a-b}\left(\frac{1}{s-a} - \frac{1}{s-b}\right)\right\} \\ &= \frac{1}{a-b} \mathcal{L}^{-1}\left\{\frac{1}{s-a} - \frac{1}{s-b}\right\} \\ &= \frac{1}{a-b}\left(\mathcal{L}^{-1}\left\{\frac{1}{s-a}\right\} - \mathcal{L}^{-1}\left\{\frac{1}{s-b}\right\}\right) \\ &= \frac{1}{a-b}(e^{at} - e^{bt}) \\ &= \frac{e^{at} - e^{bt}}{a-b} \quad (a \neq b).\end{aligned}$$

$$\text{If } a = b \text{ then } \mathcal{L}^{-1}\left\{\frac{1}{(s-a)^2}\right\} = te^{at}$$

(2)

$$\begin{aligned}\mathcal{L}^{-1}\left\{\frac{s^2}{1+s^2}\right\} &= \mathcal{L}^{-1}\left\{\frac{s^2+1-1}{s^2+1}\right\} \\ &= \mathcal{L}^{-1}\left\{1 - \frac{1}{s^2+1}\right\} \\ &= \mathcal{L}^{-1}\{1\} - \mathcal{L}^{-1}\left\{\frac{1}{s^2+1}\right\} \\ &= \delta(t) - \sin t.\end{aligned}$$

3 Ordinary differential equations